# New dichotomous implementation approach for Doppler-shift estimation and compensation in an underwater acoustic modem

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***Abstract*—As a renewed interest in ocean exploration emerges, mainly driven by a much broader autonomous presence, underwater acoustic communications have become essential. But as acoustic waves are most effective only at ranges greater than hundred meters, a number of problems arise in operational conditions, such as low throughput, long propagation delay, multi-path effect and last but not least the impact of Doppler shift. In order to compensate for Doppler effect in the decoding stage, the Doppler shift must be accurately estimated at the receiver side. Several methods can be invoked, however the use of filter banks is known to be one of the most accurate method in the state of the art. In this paper, we propose an FPGA implementation of Doppler estimation based on correlation between the received signal (which contains a known sequence) and numerous dopplerized versions of this known sequence. We take advantage of FPGA technology to run 2 comparisons in parallel, and iterate following a dichotomy approach in order to converge towards the actual Doppler value. Compared to a solution that iteratively searches for the maximum correlation with $n$ values, our approach only needs to make $\log_2 n$ comparisons, in only $\log_2 n/2$ as much time, to achieve the same precision. This makes our implementation well suited for real-time applications while still keeping the resource footprint fairly low.**

***Index Terms*—Doppler, FPGA, Underwater communications**

## I. Introduction

### A. Context

The need to explore the underwater world is crucial today for many different reasons such as security and safety, military, scientific or industrial needs. In order to be efficient in this widely unknown environment, the use of Autonomous Underwater Vehicles (AUVs) is becoming more widespread and the need for wireless communication with such drones becomes essential. But underwater communication is a real challenge because of the complexity of the communication channel. There are in fact few technologies to communicate underwater: Ultra Low Frequency radio, optical or acoustic waves [1]. Due to their versatility, acoustic modems take the lead in onboard communication equipment even though acoustic propagation is subject to scattering, multi-path and low propagation speed. Moreover, the motion of the AUV and/or the reflection on any surface wave induce a Doppler effect which needs to be estimated and compensated at the receiver side to maintain an acceptable bit error rate [2].

### B. Doppler estimation

In this context, embedded acoustic modems need to estimate Doppler shift prior to data equalization and decoding. To perform this estimation, 3 alternative processes are traditionally considered. Firstly, the use of Wide Spread Spectrum and monochromatic signal is a method which requires the emission of a Wide-Spread Spectrum Chirp (WSSC) in conjunction with a monochromatic signal. Detection is based on the WSSC with a correlator and Doppler estimation on the monochromatic signal. This has the advantage of performing a good Doppler estimation and have a fairly accurate arrival time. However, it requires a very wide bandwidth transmitter and have the possibility of non-detection [3] [4].

Secondly, another method consists in computing the signal dilation or compression. This method requires a preamble and a postamble at signal level. The time change (compression/dilation) between these signals is an estimator of the Doppler effect. It is fairly accurate and simple to process, but requires two patterns, thus reducing data rate [5]. Thirdly, the use of dedicated filter banks is a method which requires correlators arranged in a filter bank adapted to a dopplerized pattern. Since the filters are adapted, this method is highly precise in terms of Doppler and arrival time estimations, but needs high computational power consumption [6].

### C. Structure

The next section will present the use case we considered and based all presented metrics on. In section III we will present a new dichotomic approach to the filter bank that drastically reduces the number of computations required. Section IV will give the details of our hardware implementation of this dichotomic approach. Finally, some applications examples will be proposed before concluding.

## II. Use case

We consider a single carrier Phase Shift Keying (PSK) signalling with the conventional decoder depicted in Fig. 1

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and introduced in [7]. The waveform used is a composited signal which concatenate a BPSK (Binary Phase Shift Keying) header signal during 19.9 ms and a QPSK (Quaternary Phase Shift Keying) data signal during 209.7 ms. Both BPSK and QPSK have a center frequency of 27 kHz and a bandwidth of 4 kHz. The Square Root Raised Cosine (SRRC) roll-off of the SRRC filter is 0.4. We can notice that the synoptic diagram a Doppler estimation block can be decomposed in 4 steps. The signal which includes a known header is sampled and demodulated, filtered and downsampled. Finally, the result is correlated with $N_d$ dopplerized headers from $N_b$ modems.

This correlation is made by filter banks. This method consists in finding the maximum correlation peak among all tested dopplerized headers. The correlation peak is valid if its value is higher than the treshold value where the treshold is the image of the signal's energy in the analysed window. In Fig. 3 and Fig. 2, there are the trace of an orthogonal signal and a tracked signal correlated with 2 dopplerized patterns and the detection treshold in red, value nominally situated 9 dB below the autocorrelation peak, the secondary lobe of the correlation peak are 12 dB below the peak to avoïd false detection.

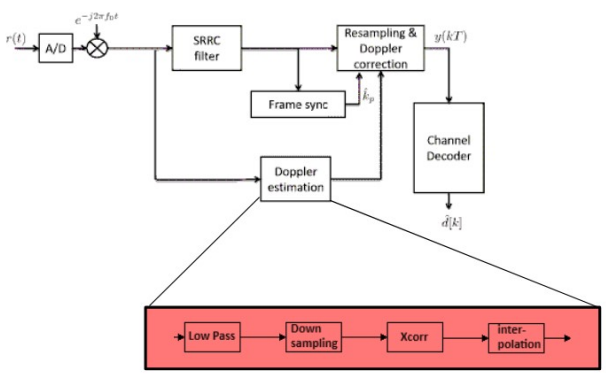


Fig. 1. Modem demodulator synoptic diagram

## III. Dichotomic approach to filter banks

Our first contribution is to organize the search space of the dopplerized version of the pattern into a binary tree.

### A. Binary Tree exploration

An example is provided in Fig. 4. In this example, we receive a communication signal with a Doppler value of 44.92 Hz *ie* 2.5 ms$^{-1}$. During the first stage of the tree, no correlation peak is detected between the received signal and the dopplerized versions (-71.8 Hz and 71.8 Hz ie. 4 ms$^{-1}$ and $-4$ ms$^{-1}$) of the known pattern. Thus, we continue the exploration by testing the whole second stage. During this stage, a correlation peak is detected between the signal and the 35.95 Hz ie. 2 ms$^{-1}$ Doppler version of the pattern. This means that we can automatically skip all the nodes that are connected to those that did not present a correlation peak. In the case there are 2 or more correlation peaks detected during a single stage, only the one with the maximum value is kept while the others are rejected. As a result, during the third stage, only the children nodes of the 35.95 Hz ie. 2 ms$^{-1}$one have to be tested. This process is reiterated until reaching the end of the binary tree. Finally, the Doppler value that is eventually kept is the one that has the maximum correlation peak between all the nodes that were not previously rejected.

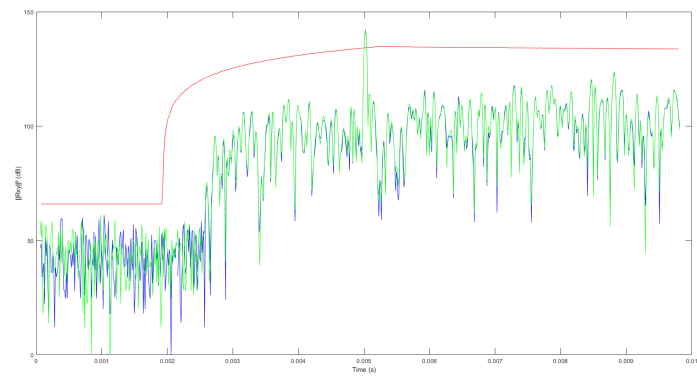

Fig. 2. results of a couple of dopplerized pattern correlated with a tracked beacon compared to the threshold

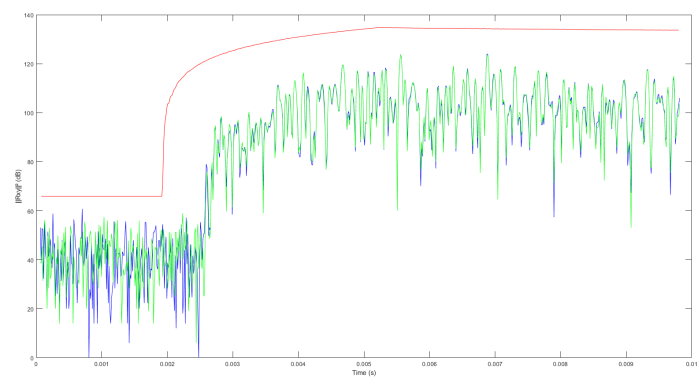

Fig. 3. results of a couple of dopplerized pattern correlated with a non tracked beacon compared to the threshold

### B. Complexity gain

This method allows to drastically reduce the number of computations required to converge towards the actual Doppler value of the signal. In the considered example, we discriminated between 14 Doppler values after only 8 actual correlation computations. The best case scenario is when a correlation peak is detected from the very first stage, as it allows to cut down half the tree at every stage. In that case only 2 computations are required each stage. Thus, after N stages, we are able to discriminate between $2^{N+1}-2$ Doppler values, after only $2N$ actual computations.

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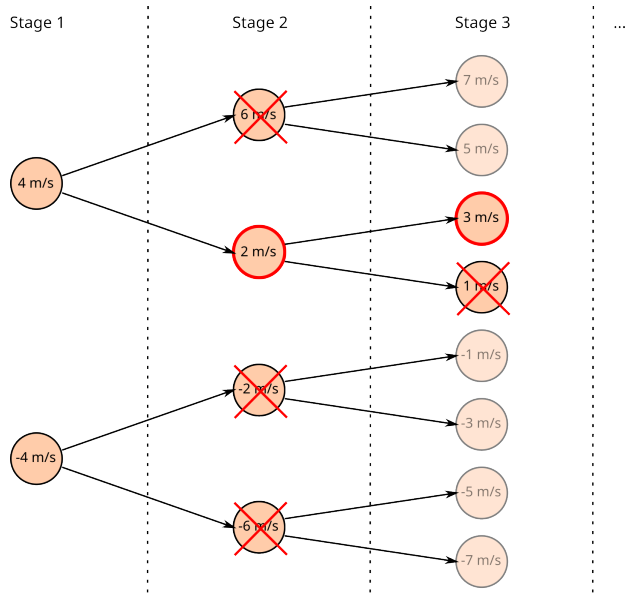


Fig. 4. Binary tree organisation of the Doppler estimation

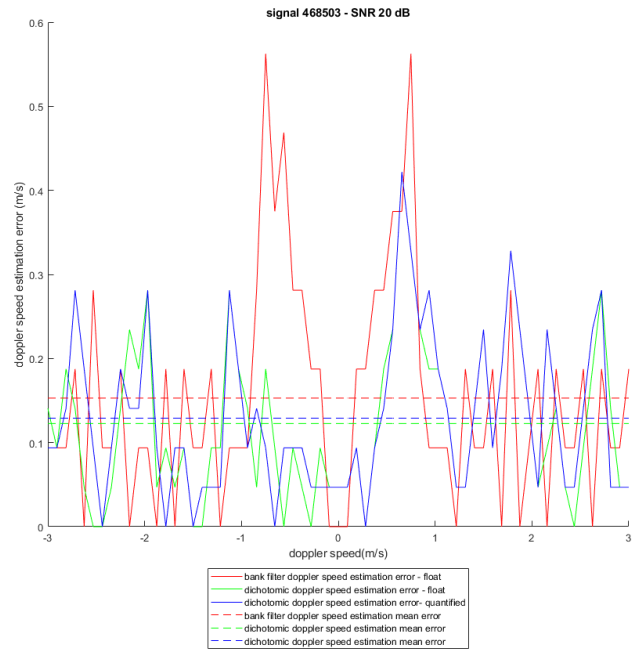


Fig. 5. Doppler estimation error at 20 dB SNR

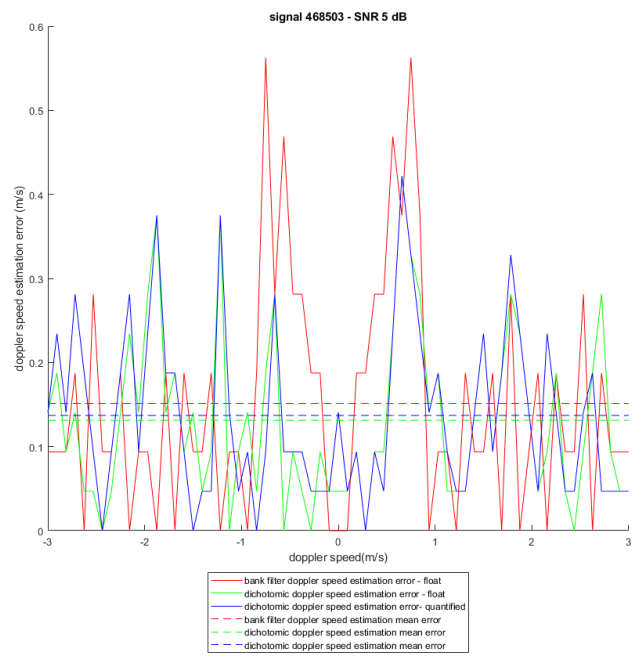


Fig. 6. Doppler estimation error at 5 dB SNR

### C. Estimation accuracy

In the case where there is one tracking signal, in simulation, we can see in Fig. 5 and 6 the error in the Doppler estimation (speed of the source) and the mean of the error for a -53.89 Hz to 53.89 Hz ie. $-3$ to 3 $\text{ms}^{-1}$and 6 stages of dichotomic structures (resolution of 1.69 Hz ie $0.09375$ $\text{ms}^{-1}$) for an Signal to Noise Ratio (SNR) of 20 dB and 5 dB. We can see that bank filter method is the worst in terms of mean error, and the dichotomic structure in fixed point is a bit worst than its floating point version.

## IV. Parallel Implementation

### A. Overview

Our second contribution is to take advantage of parallel computing architectures, such as Field Programmable Gate Arrays (FPGAs), to increase the number of dopplerized version that we can test per second. Our hardware implementation is presented in Fig. 7.

The parallel property of FPGAs means that the demodulation, filtering and downsampling steps of new samples can be performed in parallel of the correlation steps of previous sample chunks. In this implementation, the correlation part is duplicated so that we can test between two dopplerized versions of the pattern at the same time. This means that, in the best case scenario (which is if a correlation peak is detected at the first stage of the binary tree), we can go down one stage in the binary tree in the time required to compute one correlation. In this case, performing $N$ iterations is equivalent to reaching the $N^{\text{th}}$ stage of the binary tree, which is equivalent to having discriminated between $2^{N+1} - 2$ dopplerized patterns.

### B. Sliding window

One problem that can arise when using the filter banks method is if a pattern starts towards the end of a data chunk.

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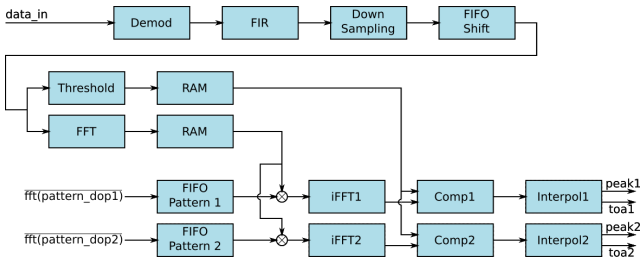


Fig. 7. Hardware implementation of the Doppler estimation chain (Data path only)

In that case, one part of the pattern will be in a data chunk while the other part will be in the next data chunk, most likely leading to a miss during the correlation computation.

To prevent this from happening, we use a sliding window, so that instead of beginning a new data chunk at the end of the previous one, we start it from the last samples of the previous data chunk.

For example, in our use case, our pattern is 226 samples long, and data chunks are 1024 samples long. We begin to treat a new data chunk every $1024 - 226$ samples, so that we are sure that the pattern is entirely in one data chunk.

In order to implement the sliding window, we use a conventional First-In-First-Out storage structure (FIFO) consisting of a dual port RAM and two counters representing the read and write addresses of the RAM. We added a shift control signal that, when activated, forces the read pointer to go back a fixed amount (this amount is equal to the size of the slide). This shift signal is activated once every time the Fast Fourier Transform (FFT) has finished fetching 1024 samples.

### C. Throughput

In our use case, the slowest component in our correlation chain is the 1024 points FFT/iFFT. It uses an iterative implementation that uses a single radix-2 butterfly operator, which can perform a computation every 2 clock cycles. As there are $N \log_2 n/2$ butterfly operations for an $N$ points FFT, and considering our implementation requires $N$ clock cycles to fetch its $N$ input points as well as $N$ clock cycles to transmit its $N$ output points, an entire FFT requires $N(2 + \log_2 N)$ clock cycles.

As our FPGA is clocked at 100 MHz, we are able to compute a new 1024 point iFFT every 122.88µs. As explained in the previous section, we begin to treat a new data chunk every $1024 - 226 = 798$ samples. Since the sampling frequency at the output of the downsampler is 9.766 kHz, we need to treat a new data chunk every 81.7ms.

As a result, we are able to compute 634 iFFTs for each data chunk. In term of complexity, this is meaning that in the best case scenario (that is, if a correlation peak is detected during the very first stage), we can discriminate between $2^{635} - 2$ patterns in real time instead of 1268 with non-optimized processing. The real gain depends of the use case.

### D. Resources utilization

The FPGA resources used by our implementation are shown in table I. The FPGA used for this is the Xilinx Zynq xc7z020clg400, and for this use case, all data signals are 16 bits wide, except for complex values, which are 32 bits wide (16 bits for the real part and 16 bits for the imaginary part).

| Component | LUTs | Flip-Flops | BRAMs | DSPs |
|---|---|---|---|---|
| Demodulation | 71 | 36 | 0 | 2 |
| FIR | 381 | 1060 | 0 | 2 |
| FIFO_shift | 108 | 40 | 1 | 0 |
| FFT | 5558 | 316 | 6 | 4 |
| iFFT (x2) | 5525 | 296 | 6 | 4 |
| Others | 76 | 123 | 2 | 6 |
| **Total** | **17244** | **2167** | **21** | **22** |

TABLE I
FPGA RESOURCES UTILIZATION

## V. Applications

The use case of this type of structure is related to underwater acoustic communications or underwater positioning (LBL or USBL mostly). In communication, the Doppler speed creates a rotation of the PSK constellation that we must compensate by Doppler shift estimation. In the same way, in positioning systems, this kind of structure estimated the most compliant Doppler shift so the real signal and the interpolation give the most relevant timing between sent and received signals. If we choose orthogonal signals as different sources, we can locate or communicate with a lot of beacons at the same time, due to great number of cross correlation. In a standard commercial systems, a 10 beacons simultaneous tracking is common, 20 or 30 beacons at the same time is unusual.

Moreover, as described in [8] the estimation of the Doppler shift effect can be use in an opportunistic way to estimate the AUV speed when decoding communications. Another use case of the Doppler shift estimation is to synchronize mobile underwater sensor networks [9].

## VI. Conclusion

In conclusion, we have developed an improvement to the filter bank method, that combines a binary tree approach with a parallel implementation in order to greatly reduce the computational resources required. However, further work is required as the reduction obtained is highly dependant on the stage of the binary tree during which a first correlation peak is observed, which makes it non deterministic. Also, it must be noted that our implementation has not been tested in a real environment yet in our platform IROMI-LMAIR [10]. It appears that this method is more accurate than bank filter method to estimate Doppler shift. The development of a dichotomic controller to scan first beacons in line of sight could be interesting.

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